Title: Training Module on Digital Manufacturing: Automated Conveyor Systems with Sensor-Based Sorting, Robotic Handling, and AGV Integration”

Background

In today’s manufacturing industry, the shift from traditional production lines to digitally integrated smart factories has transformed how material handling and process automation are carried out. Conventional conveyor systems, though reliable, lacked adaptability and intelligence. With the rise of Industry 4.0, conveyors are now embedded with sensors, robotics, and real-time monitoring systems that enable greater efficiency, flexibility, and responsiveness.

The use of IR sensors for detection and counting, color sensors for intelligent sorting, and Automated Guided Vehicles (AGVs) for transport demonstrates the transition toward cyber-physical systems where hardware and software interact seamlessly as shown in Figure 1. This training module builds on these concepts by giving learners practical exposure to automated conveyor systems integrated with robotics and AGVs, mirroring real-world applications in smart factories, warehouses, and digital manufacturing environments.

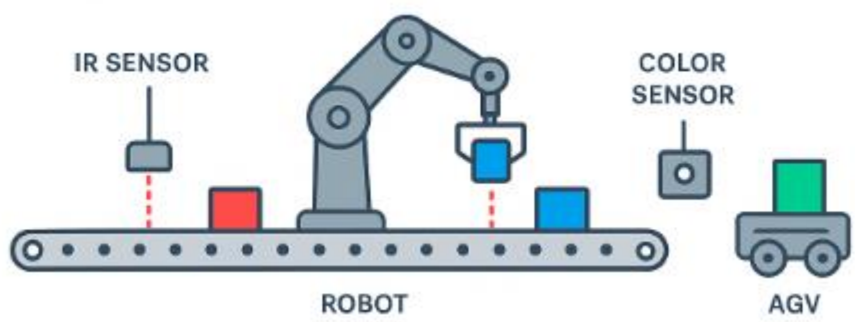

Figure 1: Digital Conveyor System

Module Overview

This training module introduces students to digital manufacturing concepts through a hands-on setup combining a straight conveyor, IR sensors, robotic arms, and Automated Guided Vehicles (AGVs). The system demonstrates how sensors enable real-time detection and counting, how robots perform intelligent part segregation based on color, and how AGVs provide flexible transport. Together, these elements replicate an Industry

4.0 environment, where automation, smart decision-making, and digital integration enhance efficiency, accuracy, and adaptability in modern material handling systems.

Learning Objectives

- Explain the basic principles of conveyor-based material handling.
- Calculate delivery time, flow rate, and loading/unloading constraints for single-direction conveyors.
- Understand how straight conveyors, IR sensors, robots, and AGVs work together in automated material handling.
- Explain part detection, counting, and color-based segregation using sensors and robots.
- Describe the role of AGVs in flexible transport and Industry 4.0 environments.
- Analyze system performance through throughput, efficiency, and bottleneck identification.

Theory

Broadly, conveyors can be classified into single-direction (straight) conveyors and continuous loop conveyors. In a single-direction conveyor, materials move along a straight path from a loading station to an unloading or processing station. Once the material reaches its destination, the conveyor does not return the load automatically; instead, the system is designed for one-way transfer. This type of conveyor is simpler, cost-effective, and well suited for linear production flows such as assembly lines, packaging, or segregation of materials, as is the case in the present lab setup. In contrast, continuous loop conveyors operate in a closed circuit where carriers move in a cycle, transporting materials on the forward trip and returning empty. These are common in large-scale production systems such as automotive plants where parts must circulate continuously between machining and assembly areas.

In the present setup, the conveyor is straight and single-directional. It is equipped with two IR sensors and integrated with robotic arms and AGVs. The first sensor, located at the initial workstation, is responsible for detecting and counting every incoming part. This allows operators to determine the exact inflow rate of materials and verify that the conveyor is operating within its designed capacity. The second IR sensor, installed at the segregation station, acts as a trigger for sorting operations. It detects the presence of each part before it undergoes color identification and robotic handling.

The segregation of parts is based on color, with three distinct categories: red, green, and blue. Once a part reaches the second sensor, a color detection system identifies its type. Blue parts are automatically picked by Robot-1, which transfers them onto an Automated

Guided Vehicle (AGV) for delivery. Similarly, green parts are picked by Robot-2 and also loaded onto an AGV. Red parts, on the other hand, are not handled by the robots; they continue along the conveyor for alternative processing or manual collection. This setup demonstrates how conveyors can be integrated with intelligent decision-making systems to create an efficient sorting mechanism.

The final element in this system is the AGV, which collects the segregated parts and delivers them to predefined storage or assembly stations. The AGV ensures mobility and flexibility in material handling, eliminating the need for fixed transport systems like forklifts or overhead cranes. By combining conveyors, robots, sensors, and AGVs, the system replicates a real-world Industry 4.0 environment where parts are not only moved automatically but also tracked, identified, and directed with minimal human intervention. Overall, this integrated material handling system highlights the synergy between mechanical transport (conveyor), intelligent detection (IR sensors), selective manipulation (robots), and autonomous delivery (AGVs). The setup allows students to study not just the working of conveyors, but also the broader interaction between sensing, decision-making, and automation in modern manufacturing systems.

Conveyor

A conveyor system is a mechanical handling device used to move materials efficiently from one point to another. They are widely used in manufacturing, warehousing, and logistics for automation, safety, and throughput improvement.

Types of Conveyors:

Single-Direction Conveyor

A single-direction conveyor is the most basic type of material handling system. It consists of a belt or chain moving in one direction to transfer items from a loading point to an unloading point as shown in Figure 2.

Example: Roller conveyor from production → assembly line.

Figure 2: Single-direction conveyor

Continuous Loop Conveyor

A continuous loop conveyor forms a closed path where carriers or pallets circulate continuously. Materials circulate in a closed loop (forward trip loaded, return trip empty). This ensures uninterrupted movement of loads, often in circular or oval layouts.

Example: Overhead trolley conveyor in automotive plants.

Figure 3: Continuous loop conveyor

Analysis

Single-Direction Conveyor

Delivery Time (Td): This represents how long it takes for a load to move between two stations. It depends directly on the distance traveled and inversely on the conveyor velocity. A higher velocity reduces delivery time but may increase wear and energy consumption.

$$Td = Ld / vc$$

- L_d = length between load & unload stations
- v_c = conveyor velocity

Flow Rate (Rf): Defined as the number of parts delivered per unit time, it reflects the efficiency of the conveyor in continuous production. Flow rate depends not only on conveyor speed but also on the spacing between loads.

$$Rf = vc / sc\ (parts/min)$$

s_c = spacing between loads

Constraint: To ensure smooth operation, the unloading time must not exceed the available loading time. Otherwise, congestion, bottlenecks, or idle times may occur.

$TU \leq TL$ (unloading time must not exceed loading time)

Continuous Loop Conveyor

Total Conveyor Length (L): This is the sum of the forward (delivery) and return paths, giving the complete loop dimension.

$$L = Ld + Le$$

Cycle Time (Tc): It represents the time required for a single carrier to complete one loop. Cycle time is influenced by the conveyor length and its speed

$$Tc = L / vc$$

Number of Carriers (nc): Carriers are evenly distributed across the conveyor loop. Their number depends on spacing and total conveyor length. More carriers increase throughput but also require careful synchronization.

$$nc = L / sc$$

Total Parts in System: The total work-in-process (WIP) on the conveyor is determined by the number of carriers and parts per carrier. This directly impacts production balance and inventory levels.

$$Parts = nc \times (Ld / L) \times np$$

Flow Rate:

$$Rf = (np \times vc) / sc$$

Class Example

Scenario: A straight conveyor is 35 m long and moves at 40 m/min. Tote pans (each holding 20 parts) are loaded at the entry station every 25 seconds. Determine (1) the spacing between consecutive pans, (2) the maximum possible flow rate in parts/min, and (3) the maximum allowable unloading time so that no congestion occurs at the unload station.

Step 1: Delivery time and unit conversions:
Delivery time helps sanity-check: $Tc = L / vc$ = 35 / 40 = 0.875 min = 52.5 s.
Also convert the loading interval to minutes: time for loading ($t - load$) = 25 s = 25/60 = 0.4167 min.

Step 2: Spacing between pans (sc):
Loads introduced every t_load minutes on a conveyor moving at v (m/min) yield nominal spacing $sc = v \times t - load$
Compute: sc = 40 × 0.4167 ≈ 16.67 m.

Step 3: Maximum possible flow rate (Rf):
Totes per minute = 60 / t-load(sec) = 60 / 25 = 2.4 totes/min.
Each tote carries 20 parts → Rf = 2.4 × 20 = 48 parts/min.

Step 4: Constraint on unloading time (T-U):
To avoid backlog at the unload station, unloading must not be slower than loading cadence: T-U ≤ T_L.
Here T-L = t-load = 25 s, hence T-U,max = 25 s.

Interpretation:
- Spacing ≈ 16.67 m keeps totes evenly distributed at the given speed.
- Ideal line capacity ≈ 48 parts/min (ignoring downtime/inefficiencies).
- Any T-U > 25 s creates a queue; T-U < 25 s provides buffer.

Activities & Exercises

Activity 1: Quick Calculation

A conveyor is 50 m long, moving at 25 m/min. Parts are loaded every 0.5 min.
- Compute delivery time.
- Compute max flow rate.

Activity 2: System Evaluation

A loop conveyor has total length 240 m, velocity 30 m/min, carrier spacing 10 m, each holds 3 parts.
- How many carriers are in the loop?
- What is the total capacity of the system?

Activity 3: Case Study

Compare two designs for moving parts between machining and assembly:
- Design A: Single-direction conveyor, 20 m, 30 m/min, spacing 1.5 m.
- Design B: Continuous loop conveyor, 50 m total length, 40 m/min, carrier spacing 2 m, 2 parts per carrier.

Which design is more efficient? Discuss trade-offs in space, throughput, and cost.

Exercise

Design a single-direction conveyor of 50 m moving at 25 m/min, spacing = 2 m per load.

- Calculate delivery time.
- Calculate flow rate.
- Check if unloading (20 sec) ≤ loading (25 sec).

Assessment Questions

Question: Define delivery time for a single-direction conveyor.

Question: Why must unloading time be ≤ loading time? What happens otherwise?

Question: A continuous loop conveyor is 200 m long, moving at 50 m/min, with 10 m spacing and carriers holding 4 parts. Find:

a) Number of carriers

b) Total parts in system

c) Maximum flow rate

Question: List three advantages and disadvantages of continuous loop conveyors vs single-direction conveyors.

Question: Explain how conveyor design impacts production line balancing.

Need for Shift from Conventional to Digital Conveyor Systems

Conventional conveyor systems, while reliable for basic material transport, face limitations in flexibility, efficiency, and real-time monitoring. They usually operate in isolation, without connectivity to sensors, data systems, or decision-making tools, making it difficult to adapt to dynamic production requirements. In contrast, digital conveyor systems integrate IoT devices, sensors, and data analytics to provide real-time visibility of material flow, predictive maintenance, and adaptive control. This shift is essential for aligning manufacturing with Industry 4.0 standards, where automation, transparency, and smart decision-making are critical. Moving towards digital conveyors not only reduces downtime and manual intervention but also enhances productivity, traceability, and overall system intelligence.

Training Activity

The aim of training activity is to introduces students to digital manufacturing concepts through a hands-on setup combining a straight conveyor, IR sensors, robotic arms, and Automated Guided Vehicles (AGVs).

Training Delivery Framework

- Lecture: 30 minutes (concepts & formulas)
- Worked Examples: 20 minutes
- Hands-on Exercises: 30 minutes (students solve in groups)
- Case Study & Discussion: 20 minutes
- Assessment Quiz: 20 minutes

Equipment

The following equipment is used in this training module:
- Motor-driven conveyor belt (straight, single-direction).
- IR Sensor-1: Part counting at Station-1.
- IR Sensor-2: Trigger for segregation at Station-2.
- Color identification system (camera/vision or pre-coded parts).
- Robot-1: Picks blue parts and places them on AGV.
- Robot-2: Picks green parts and places them on AGV.
- Automated Guided Vehicle (AGV) prototype for transporting segregated parts.
- Control system (Arduino / Raspberry Pi / PLC).

Procedure

1. Switch on the conveyor and allow parts of three colors (red, green, blue) to flow.
2. At Station-1, IR Sensor-1 detects and counts each part. Record the total number of incoming parts.
3. At Station-2, IR Sensor-2 detects the part before segregation. The color detection system identifies each part:
   - Blue part → picked by Robot-1 and placed on AGV.
   - Green part → picked by Robot-2 and placed on AGV.
   - Red part → bypasses the robots and continues forward.
4. AGV transports segregated parts to designated storage or assembly stations.
5. Repeat the cycle and collect data on:
   - Total number of parts counted.
   - Number of each color segregated.
   - Robot cycle times and AGV turnaround time.
   - Segregation efficiency and overall throughput.

Class Example

In this training lab, a straight conveyor of 10 meters length runs at a speed of 30 meters per minute. At the entry point, the first infrared (IR1) sensor detects incoming parts, which arrive at a rate of one part every 2.5 seconds. These parts are of three different colors, with a distribution of approximately 40% blue, 35% green, and 25% red.

As each part reaches the second sensor (IR2) at the segregation station, the system identifies its color and directs it accordingly. Robot-1, positioned at this station, is dedicated to handling blue parts. It operates with a cycle time of 5 seconds per pick-and-place action, giving it the capacity to process about 12 parts per minute. Robot-2, responsible for handling green parts, works at a cycle time of 6 seconds, corresponding to a capacity of roughly 10 parts per minute. Red parts are not picked by the robots and instead continue further along the conveyor line for manual handling or other processes.

The segregated blue and green parts are then transferred onto an Automated Guided Vehicle (AGV). Each AGV in the system can carry 25 parts per trip, and requires about 4 minutes to complete a round-trip delivery from the segregation station to the storage area and back.

Given Data
- Conveyor: length $L = 10$ m, speed $v = 30$ m/min
- Arrival cadence at Station-1 (IR1): one part every 2.5 s
- Color mix at Station-2 (IR2): 40% blue, 35% green, 25% red
- Robot-1 (Blue): cycle time $Tc1 = 5.0$ s → capacity = 12 parts/min

- Robot-2 (Green): cycle time $T_{c2} = 6.0$ s → capacity = 10 parts/min
- AGV: capacity $C = 25$ parts/trip, round-trip time $T_{trip} = 4$ min

Solution

Step 1: Delivery time to Station-2
$T_d = L / v = 10 / 30 = 0.333$ min = 20 s
A part detected by IR1 reaches IR2 about 20 s later.

Step 2: Inflow rate at IR1 (Station-1)
$R_{in} = 60 / 2.5 = 24$ parts/min.
So per minute: 24 parts arrive onto the conveyor.

Step 3: Color-wise arrivals at Station-2
$R_{blue} = 0.40 \times 24 = 9.6$ parts/min
$R_{green} = 0.35 \times 24 = 8.4$ parts/min
$R_{red} = 0.25 \times 24 = 6.0$ parts/min
(Red bypasses robots; blue goes to Robot-1, green to Robot-2).

Step 4: Robot capacities and utilizations
Blue robot: capacity = 12 parts/min, utilization = 9.6 / 12 = 80%
Green robot: capacity = 10 parts/min, utilization = 8.4 / 10 = 84%
Both robots can keep up with demand.
Total picked to AGV buffer = 18 parts/min.

Step 5: AGV transport capacity and bottleneck check
$R_{AGV} = C / T_{trip} = 25 / 4 = 6.25$ parts/min
Compare: 18 picked/min > 6.25 shipped/min → AGV is bottleneck.
Backlog growth = 11.75 parts/min (~118 parts over 10 minutes).

Step 6: How many AGVs are needed?
$n = R_{picked} / R_{AGV} = 18 / 6.25 = 2.88$ → need 3 AGVs.
With 3 AGVs: capacity = 18.75 parts/min ≈ matches 18 picked/min.
Alternative: if only 1 AGV is used, trip time must be ≤ 1.39 min, or capacity ≥ 72 parts/trip (impractical).

Step 7: Quick 10-minute run summary
IR1 count: 240 parts
Robots picked: 180 parts
1 AGV shipped: ~63 parts, buffer grows to ~117 parts
3 AGVs shipped: 180 parts, no buffer build-up.

Conclusion
This example shows that in a straight conveyor + sensor + robot + AGV system, the AGV loop can become the bottleneck even when robots are within their capacity. Matching transport resources with robot output is critical to prevent WIP build-up.

Activity 1: Robot Bottleneck Check

Scenario

In the lab, a straight conveyor of 8 m length moves at 20 m/min. Parts arrive at IR1 every 3 seconds. The color distribution is 50% blue, 30% green, and 20% red. At the segregation station, Robot-1 handles blue parts with a cycle time of 4.5 seconds, while Robot-2 handles green parts with a cycle time of 7 seconds. Red parts bypass the robots.

Tasks for Students

1. Calculate the delivery time from IR1 to IR2.
2. Determine the total inflow rate of parts per minute.
3. Calculate the blue and green part arrival rates per minute.
4. Compare these arrival rates with the robot capacities (12 parts/min for Robot-1, ~8.6 parts/min for Robot-2).
5. Identify if either robot becomes a bottleneck, and explain the expected effect on the system.

Activity 2: AGV Capacity Balancing

Scenario

A conveyor of 12 m length runs at 25 m/min, delivering one part every 2 seconds. The mix of parts is 30% blue, 45% green, and 25% red. Robot-1 (Blue) has a cycle time of 5 s, and Robot-2 (Green) has a cycle time of 6 s. Both robots place segregated parts onto an AGV.

The AGV can carry 20 parts per trip and requires 5 minutes per round trip to storage and back.

Tasks for Students

1. Compute the total arrival rate of parts at IR1 per minute.
2. Find the expected blue and green inflows per minute.

3. Calculate the robot utilizations (are they under or over capacity?).
4. Determine how many parts the robots send to the AGV buffer each minute.
5. Compare this with the AGV capacity (20/5 = 4 parts/min).
6. Decide how many AGVs would be needed to balance the system, and justify the answer.

Assessment Questions

1. Why is IR Sensor-1 necessary in this system?
2. What factors affect segregation efficiency?
3. How do robot cycle times impact throughput?
4. How does AGV turnaround time limit system capacity?

Conclusion

This training module demonstrates how conveyors, sensors, robots, and AGVs integrate into a modern automated material handling system. The straight conveyor system used in the lab highlights the principle of single-direction conveyors combined with color-based segregation and robotic handling. The inclusion of AGVs for transportation replicates an Industry 4.0 environment, where automation, detection, and intelligent decision-making work together to improve efficiency, accuracy, and throughput. Students are able to connect theoretical formulas with hands-on experience, reinforcing concepts of conveyor analysis, robotic cycle time, and system throughput in a practical setting.